# JAYA BHARATH MOTHUKURI

**AI Full Stack Engineer | Software Engineer**
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## PROFESSIONAL SUMMARY

AI Full Stack Engineer with 3+ years of experience building scalable AI-driven applications, distributed microservices, and cloud-native solutions using Python, Java, and AWS. Experienced in developing multi-agent AI systems with LangGraph, LLMs, semantic search, and RAG pipelines for intelligent automation. Strong background in designing high-performance backend services, REST APIs, and event-driven architectures using Spring Boot, Kafka, and Redis. Proven ability to optimize AI workflows by transitioning from static, keyword-based pipelines to dynamic agentic architectures. Passionate about building production-ready, scalable systems that combine AI, cloud technologies, and modern software engineering best practices.

## TECHNICAL SKILLSET

- **AI/ML & Data Science:** RAG, Vector-less RAG, Graph RAG, LangChain, LangGraph, LLM Integration (Ollama, Qwen 2.5), HuggingFace Sentence Transformers, FAISS, Semantic Search, Ensemble Learning (XGBoost, CatBoost), TensorFlow, Scikit-Learn.
- **Backend, Cloud & Infrastructure:** Node.js, Python (Flask/Django), Java, C, PostgreSQL, Middleware, AWS (Bedrock, SageMaker, Lambda, EC2, S3), Docker, Kubernetes, RESTful APIs.
- **Frontend Development:** HTML5, CSS3, JavaScript (ES6+), TypeScript, React.js, Redux, D3.js, Tailwind CSS.
- **Specialized Domains:** Web3 (DIDs, Verifiable Credentials), Data Visualization, Software Security, Digital Video Processing, Knowledge Representation.
- **Engineering & Design:** Git, Agile/Scrum, CI/CD, Figma, Android Studio, User Research, Wireframing.

## PROFESSIONAL EXPERIENCE

**Full Stack Developer | Linked Creds (Predictive UX), USA| Jan 2026 – Present**

- Architected a multi-agentic RAG pipeline for an advanced Resume Analyzer, initiating semantic embeddings via HuggingFace Sentence Transformers.
- Designed an orchestrator worker flow managing specialized AI agents (Parsing, Skill Section Detection, Skill Extraction, Normalization, and Mapping) utilizing FAISS vector stores and the O*Net taxonomy.
- Engineered an evidence binding agent with a human-in-the-loop (HITL) verification mechanism for cryptographically signing digital credentials.
- Developed a responsive React interface to manage credential lifecycles, reducing user onboarding friction by 30% through streamlined UI workflows.
- Engineered a decentralized identity layer utilizing DIDs and cryptographic signatures, improving data integrity audit speeds by 40%.

**Software Engineer | Walmart Global Tech, USA | Jun 2025 – Jan 2026**

- Scaled a distributed microservices architecture using Java, Spring Boot, and Kafka to process 2.5 million+ concurrent transactions during peak holiday retail events.
- Optimized real-time inventory services by implementing multi-level caching with Redis, reducing database read latency by 60% and improving checkout throughput.
- Engineered high-performance RESTful APIs integrated with Azure Cloud, ensuring 99.99% availability for global e-commerce search and personalization engines.

**Software Engineer | Maxgen Technologies, India | Jul 2021 – Jul 2023**

- Built microservices using Python & PostgreSQL, supporting high-concurrency traffic of 50k+ daily active users for enterprise-level web applications.
- Streamlined deployment workflows by implementing automated CI/CD pipelines, reducing manual deployment errors by 20% and cutting release cycles from 5 days to 2 days.
- Refactored legacy UI components into reusable React hooks, improving frontend rendering speed by 35% and reducing the codebase size by 15%.
- Ensured 99.9% uptime for critical services through rigorous unit testing (achieving 90% code coverage) and automated monitoring scripts.

## PUBLICATIONS & PROJECTS

**Lead Researcher | IEEE Publication | March 2023**

- Built a software maintainability prediction framework using Gradient Boost, AdaBoost, and XGBoost, optimizing model accuracy on QUES/UIMS datasets.
- Validated performance using MSE, RMSE, and Pred(0.3) metrics to enable precise maintenance effort estimation in the SDLC.

**Lead Developer | Visual Analysis of Sleep Stages | Jan 2024 – May 2024**

- Developed a visualization dashboard using Flask and D3.js to analyze Polysomnography (PSG) signals & physiological transitions.
- Translated signals into hypnograms and radial charts, enhancing diagnostic capabilities for clinical abnormalities.

**AI Developer | Low Light Video Enhancement | Aug 2023 – Dec 2023**

- Implemented DecomNet and RelightNet deep learning frameworks in TensorFlow to enhance low-light video visibility through temporal filtering.

**AI Developer | Multi-Agentic AI Blog Generator**

- Architected a multi-agentic blog writing platform utilizing an orchestrator-worker framework to dynamically coordinate specialized AI agents for drafting, editing, and formatting.
- Engineered a RAG pipeline powered by semantic search for factual grounding, and integrated an evaluator framework.

## EDUCATION

- **M.S. in Computer Science** | Arizona State University | **GPA: 3.73/4.0** | **May 2025**
- **B.Tech in CS (AIML)** | SRM University | **GPA: 8.58/10** | **July 2023**